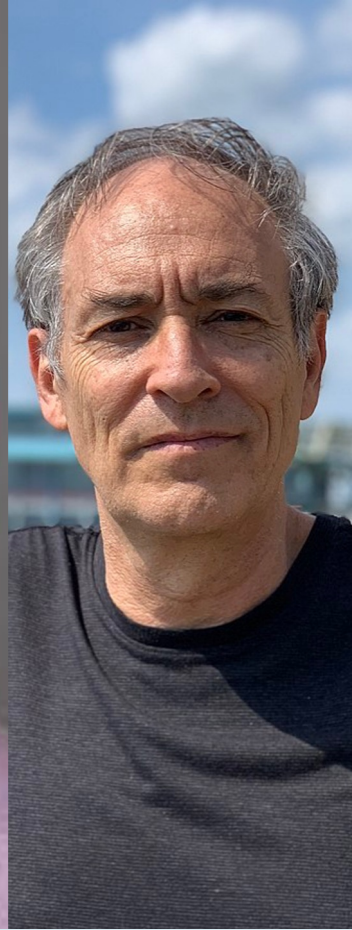
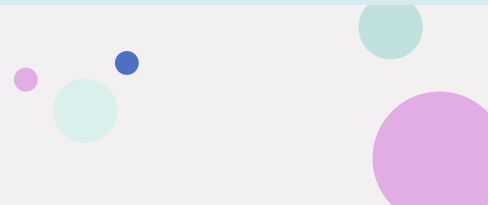




Lecture 10

MAC and Hash

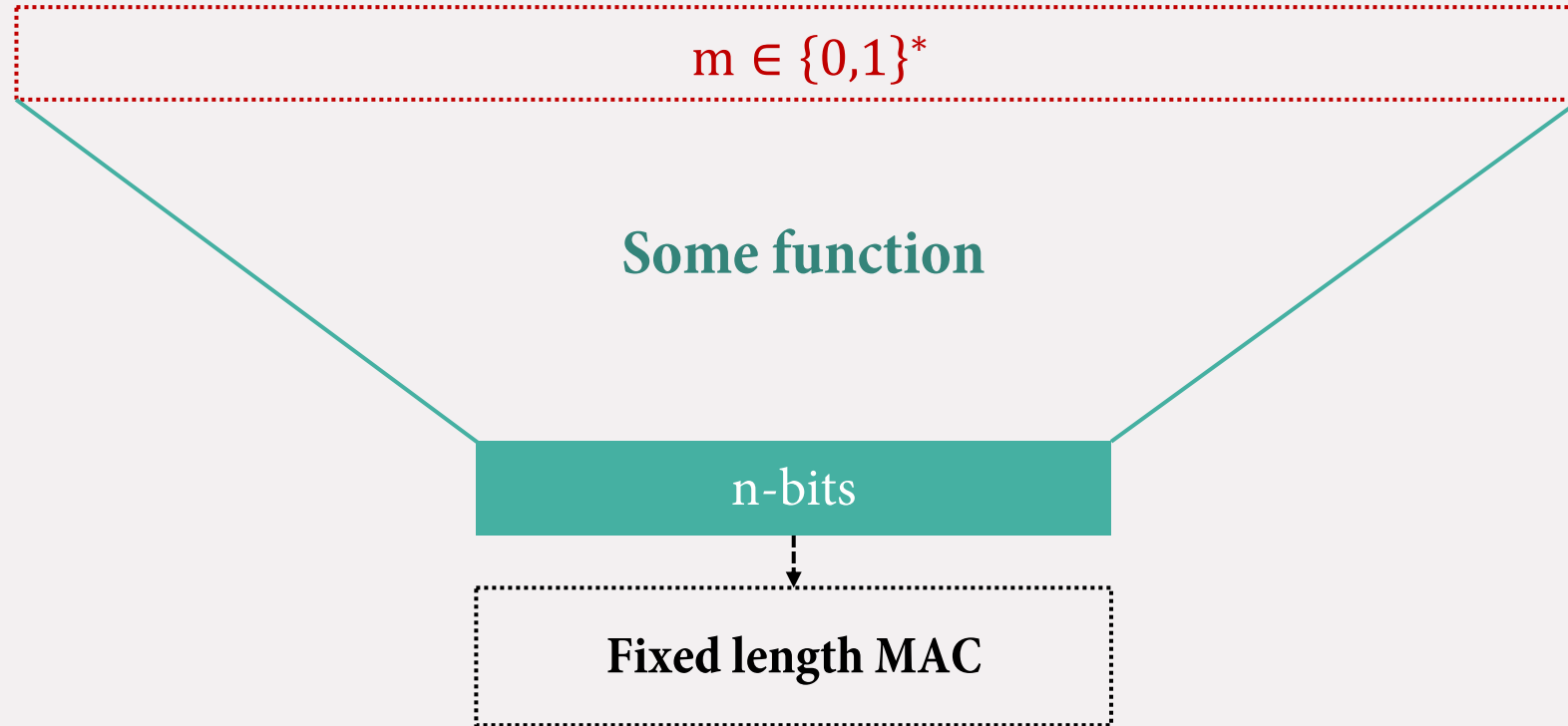
CS409: Introduction to Cryptography
Sruthi Sekar



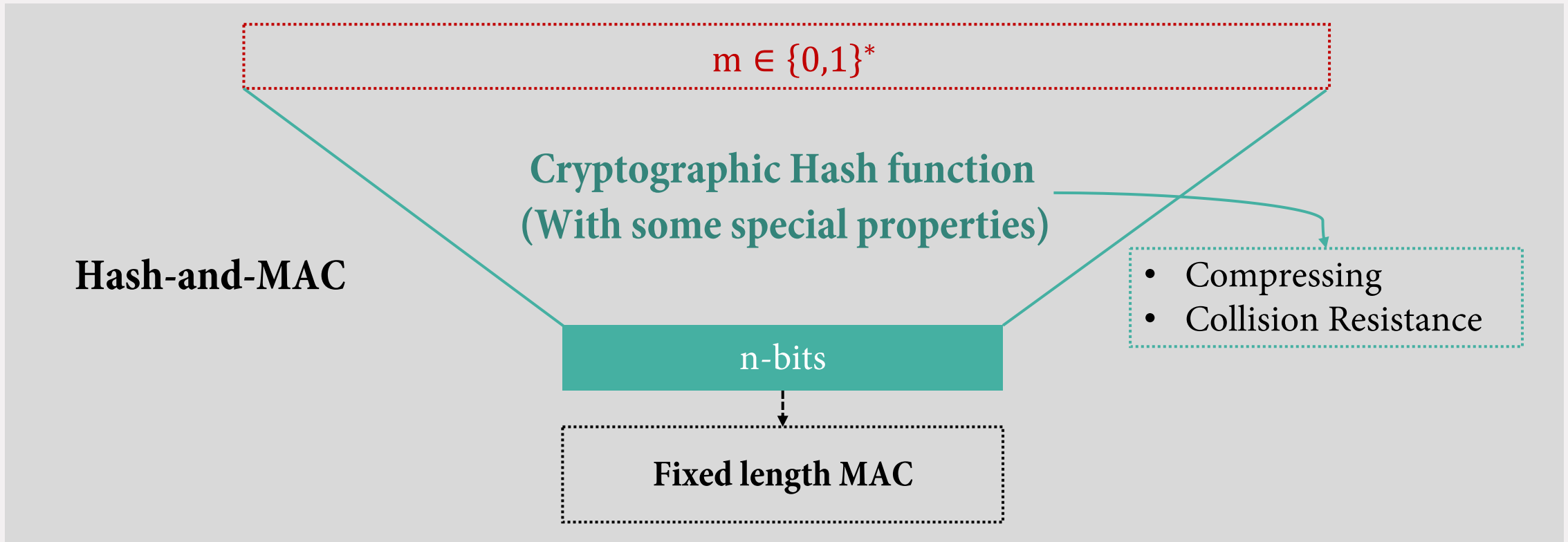
Recap

- Authenticated Encryption
(CCA Security + Ciphertext Integrity).
- Message Authentication Codes – EU-SCMA security.
- Domain extension of MACs (using PRFs)
 - Theoretical way
 - CBC-MACs
 - **Today: another way using Hash functions**

MAC: Domain Extension



MAC: Domain Extension



HMAC

In fact, we can use this new primitive to build a MAC for messages of arbitrary length without using a PRF!

Today



Cryptographic Hash Functions



Domain Extension

Merkle-Damgård Transform

HMAC

Cryptographic Hash Function

$$H: \mathcal{M} \rightarrow \mathcal{T} \text{ where } |\mathcal{M}| \gg |\mathcal{T}|$$

Collision for H

A pair of messages $m_0, m_1 \in \mathcal{M}$ such that $H(m_0) = H(m_1)$ and $m_0 \neq m_1$

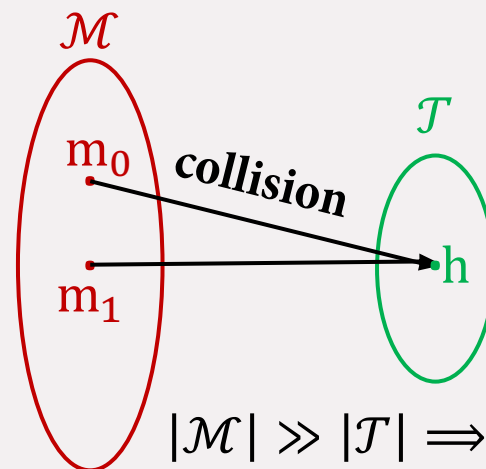


Collision Resistance

H is collision-resistant if, for all (explicit) efficient algorithms $\mathcal{A}$

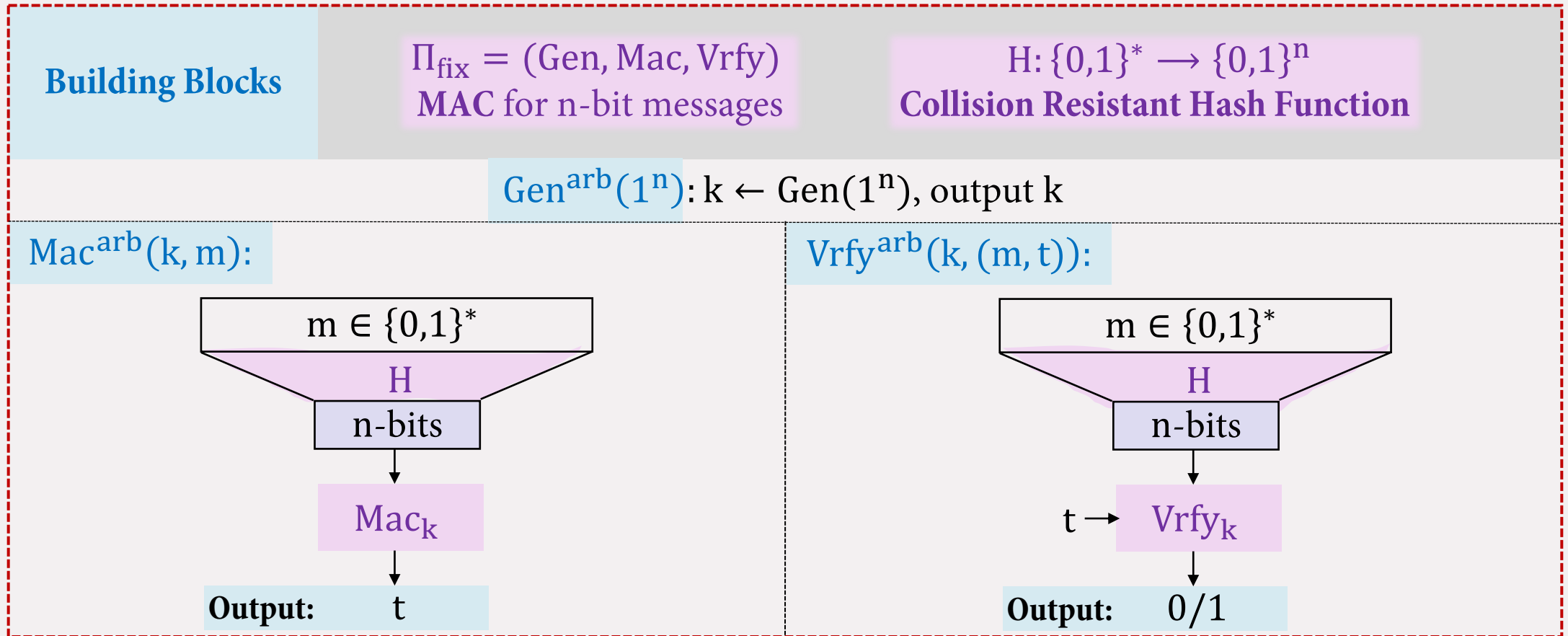
$$\text{Adv}_{\text{CR}}[\mathcal{A}, H] = \Pr[\mathcal{A} \text{ outputs collision for } H] \leq \text{negligible}$$

Example: **SHA-256** takes arbitrary long inputs and outputs 256 bits



$|\mathcal{M}| \gg |\mathcal{T}| \Rightarrow$ there will be lot of collisions

Application: Hash-and-MAC



Construction 1 (Hash-and-MAC): $\Pi_{\text{arb}} = (\text{Gen}^{\text{arb}}, \text{Mac}^{\text{arb}}, \text{Vrfy}^{\text{arb}})$ for $\mathcal{M} = \{0, 1\}^*$



Why is collision resistance of H necessary for EU-CMA security of Π_{arb} ?

If you had a collision m_0, m_1 then on getting (m_0, t_0) , **Mallory** can output (m_1, t_0) as a forgery!

Hash Functions: Theory vs Practice

Theoretical Constructions

$$\{H_k\}_{k \in \{0,1\}^n}$$

It is typically a keyed family of functions.
A Gen algorithm picks one k

There is a security parameter n

Security proved under well-studied assumptions

If you have a family of functions, you do not have the space to hardcode collisions for all functions!

Practical Constructions

E.g.: SHA3, SHA256, SHA1, MD5, MD4

Single function and not a family of functions
("unkeyed")

No security parameter knob!

Use until some finds a collision!

Not secure, theoretically! Hardcode the collision into algorithm A, it runs in constant time!

Secure: no
collision found

Broken:
collision found

Example: Collision for MD5



139KB

MD5

128 bits

5c827c0eba9cfaa647c1a489bea77c60



139KB

MD5

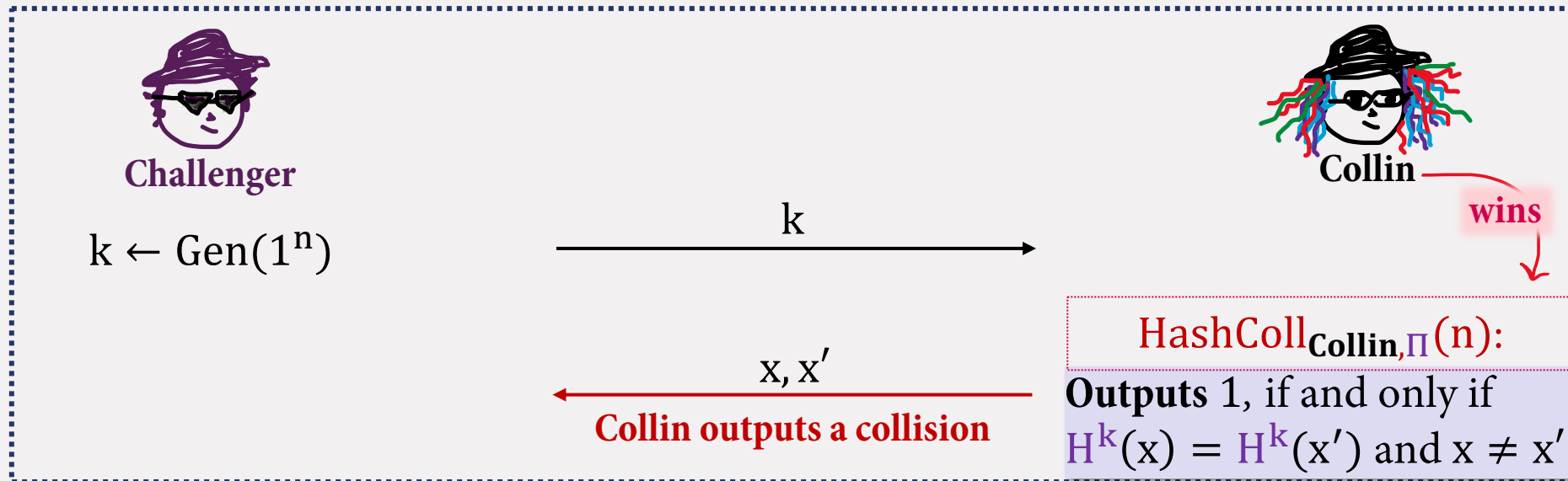
128 bits

5c827c0eba9cfaa647c1a489bea77c60

Collision Resistant Hash Functions

$\Pi = (\text{Gen}, H)$: $\text{Gen}(1^n) \rightarrow k$; H^k takes input $x \in \{0,1\}^*$ and outputs $H^k(x) \in \{0,1\}^{\ell(n)}$

If H^k only defined for $\{0,1\}^{\ell'(n)}$ for $\ell'(n) > \ell(n)$ called a **compression function**.



Definition 1

A hash function $\Pi = (\text{Gen}, H)$ is **collision resistant** if, for all **PPT** adversary **Collin**, there exists a negligible function **negl** such that, $\forall n$

$$\Pr[\text{HashColl}_{\text{Collin}, \Pi}(n) = 1] \leq \text{negl}(n)$$

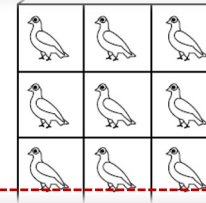
Generic Birthday Attacks

Let $H: \{0,1\}^* \rightarrow \{0,1\}^\ell$ be a hash function (key is not relevant for the attack, hence dropped in notation)

What is a trivial collision-finding attack that **Collin** can launch?

Evaluate H on $2^\ell + 1$ distinct points

Pigeonhole principle



Can you do better with the birthday paradox in mind?



Collin's strategy:

1. Choose $q = 2^{\ell/2}$ random inputs $x_1, x_2, \dots, x_q$ (distinct w.h.p)
2. Evaluate $y_i = H(x_i)$ for all $i = 1, \dots, q$.
3. Look for a collision $y_i = y_j$. If not found, go to step 1

What is the probability of finding a collision?

$\sim 1/2$

$O(2^{\ell/2})$ time and space
(expected to only take 2 iterations!)

***Treat H as a random function for this analysis. (This is alright since it can be shown that this is actually the worst case: collisions will occur with a higher probability for H that deviates from random.)**

Concretely, for T -time Collin, the hash function output must be at least $2\log T$ bits!

Today



Cryptographic Hash Functions

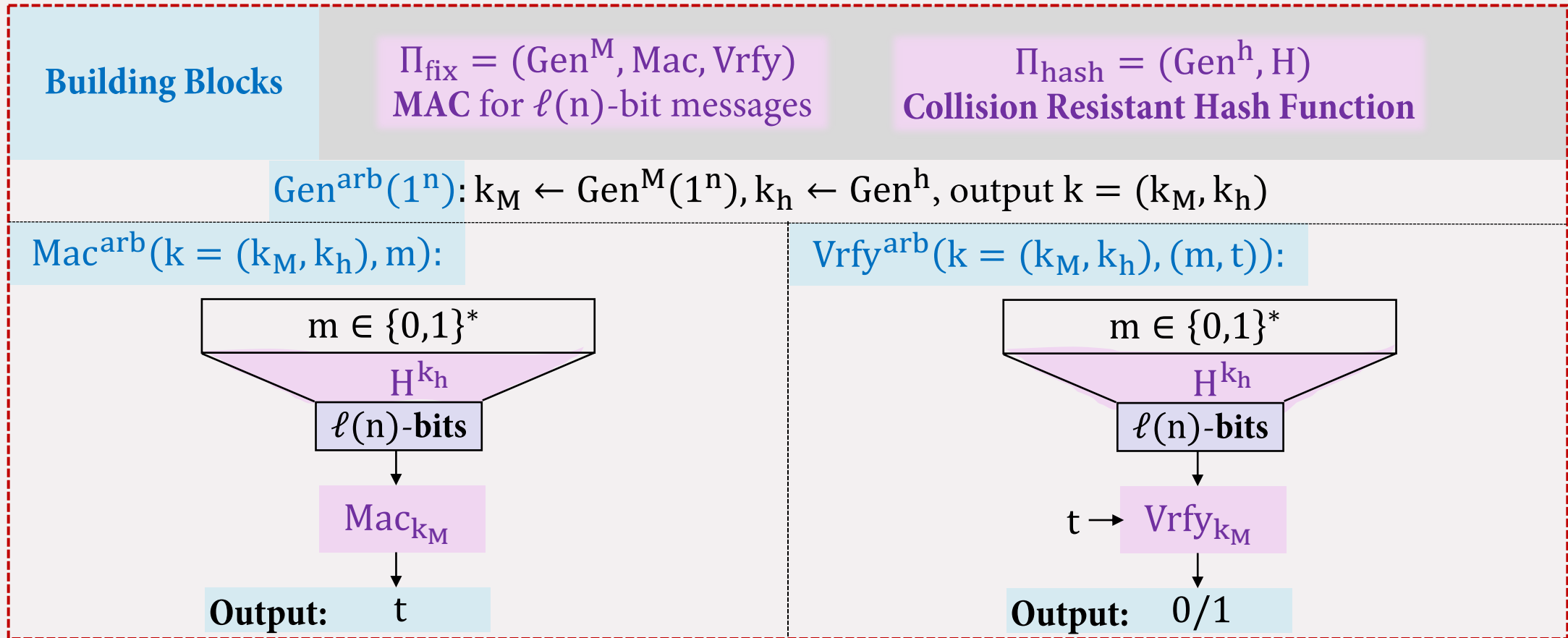


Domain Extension

Merkle-Damgård Transform

HMAC

Application: Hash-and-MAC



Construction 1 (Hash-and-MAC): $\Pi_{\text{arb}} = (\text{Gen}^{\text{arb}}, \text{Mac}^{\text{arb}}, \text{Vrfy}^{\text{arb}})$ for $\mathcal{M} = \{0, 1\}^*$

Theorem 1

If Π_{fix} is an EU-CMA-secure MAC for $\ell(n)$ -bit messages and Π_{hash} is collision resistant, then Π_{arb} is an EU-CMA-secure MAC for arbitrary-length messages.

Application: Hash-and-MAC

Theorem 1

If Π_{fix} is an EU-CMA-secure MAC for $\ell(n)$ -bit messages and Π_{hash} is collision resistant, then Π_{arb} is an EU-CMA-secure MAC for arbitrary-length messages.

Proof Sketch Suppose $\exists \text{Mallory}^{\Pi_{\text{arb}}}$ that breaks EU-CMA security of Π_{arb}



(Intuition) There are only two ways in which $\text{Mallory}^{\Pi_{\text{arb}}}$ can win captured by cases below:

Case 1 (use collision resistance of Π_{hash} to argue this only happens with negligible probability)

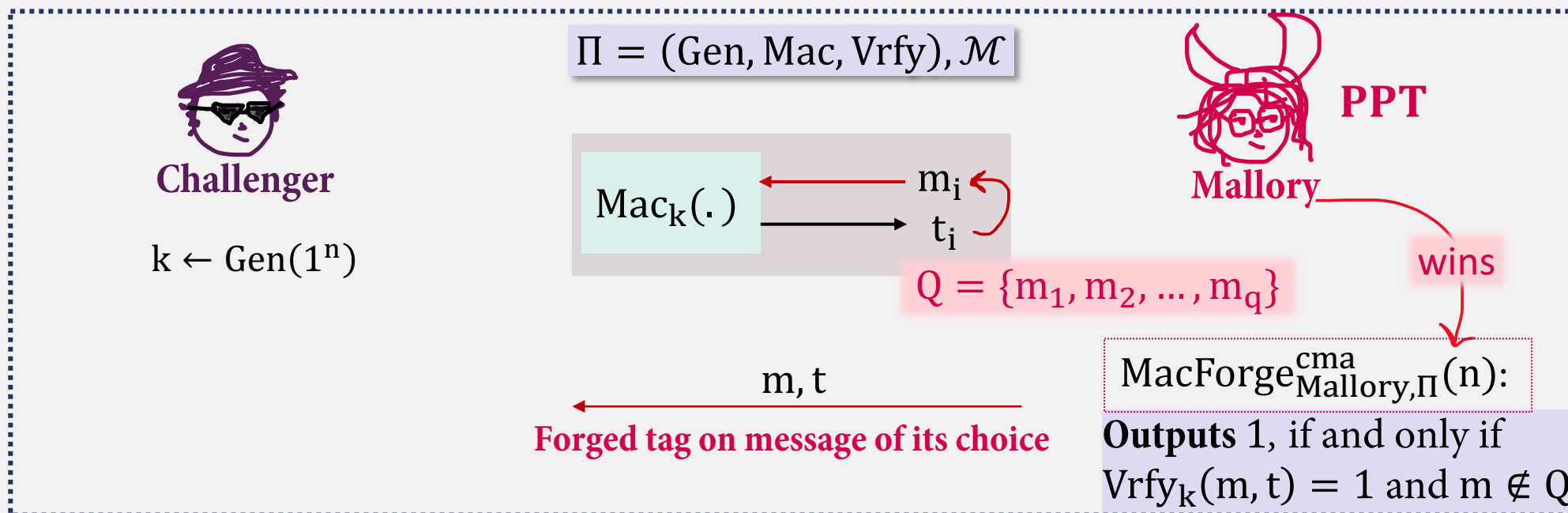
A collision (m, m') can be efficiently found ($H(m) = H(m') \wedge (m \neq m')$), used by $\text{Mallory}^{\Pi_{\text{arb}}}$:
Query $\text{Mac}_k^{\text{arb}}(.)$ on m to get t . Output the forgery (m', t) , for $m' \notin Q$

Case 2 (use EU-CMA security of Π_{fix} to argue this only happens with negligible probability)

If no such collision can be efficiently found, then the only way $\text{Mallory}^{\Pi_{\text{arb}}}$ can forge and win is by breaking the EU-CMA security of Π_{fix}

RECALL

Defining EU-CMA Security for MAC



Lecture 8, Definition 2

Π is **existential unforgeable under an adaptive chosen-message attack (EU-CMA)**, if, for all **PPT** adversary **Mallory**, there exists a negligible function **negl** such that $\forall n$

$$\Pr[\text{MacForge}_{\text{Mallory}, \Pi}^{\text{cma}}(n) = 1] \leq \text{negl}(n)$$

Application: Hash-and-MAC

Theorem 1

If Π_{fix} is an EU-CMA-secure MAC for $\ell(n)$ -bit messages and Π_{hash} is collision resistant, then Π_{arb} is an EU-CMA-secure MAC for arbitrary-length messages.

Proof Sketch Suppose $\exists \text{Mallory}^{\Pi_{\text{arb}}}$ that breaks EU-CMA security of Π_{arb}

Suppose **coll** is the event that a collision can be found efficiently.

$$\Pr \left[\text{MacForge}_{\text{Mallory}^{\Pi_{\text{arb}}, \Pi_{\text{arb}}}}^{\text{cma}}(n) = 1 \right] \leq \underbrace{\Pr[\text{coll}]}_{\text{This is negligible by collision resistance of } \Pi_{\text{hash}}} + \underbrace{\Pr \left[\text{MacForge}_{\text{Mallory}^{\Pi_{\text{arb}}, \Pi_{\text{arb}}}}^{\text{cma}}(n) = 1 \mid \overline{\text{coll}} \right]}_{\text{(Simple Reduction to EU-CMA of } \Pi_{\text{fix}} : \text{details on board)}}$$

This is negligible by collision
resistance of Π_{hash}

(Simple Reduction to EU-CMA of Π_{fix} : details on board)

Today



Cryptographic Hash Functions



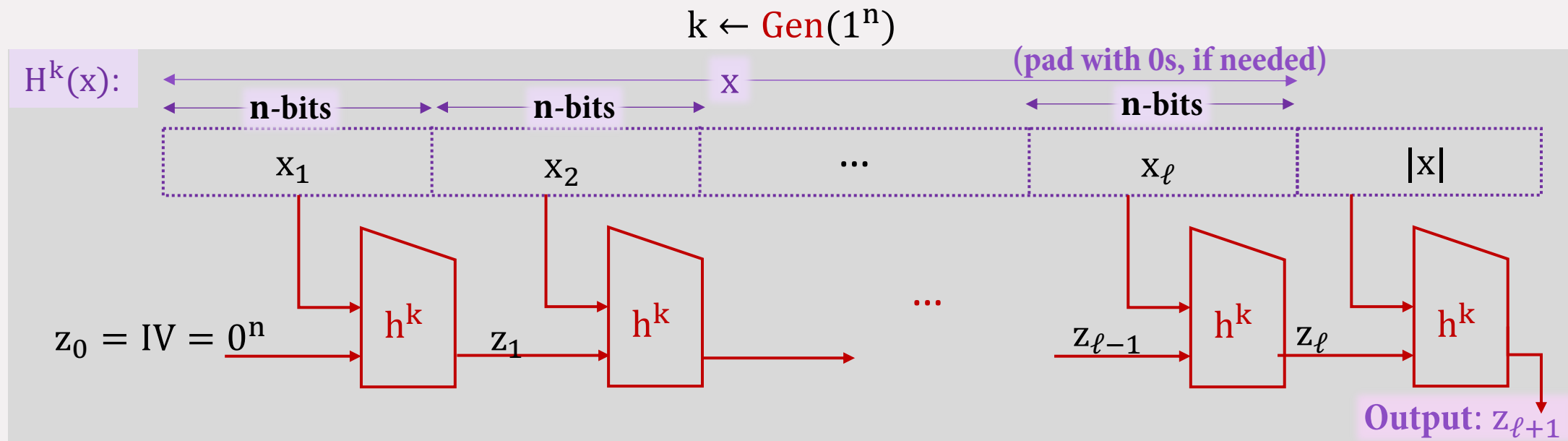
Domain Extension

Merkle-Damgård Transform

HMAC

Merkle Damgård Transform

(Gen, h) : a compression function with input length $2n$ and output length n



Construction 2

Theorem 2



[Katz and Lindell, second edition, Theorem 5.4]

If (Gen, h) is collision resistant, then so is (Gen, H) .

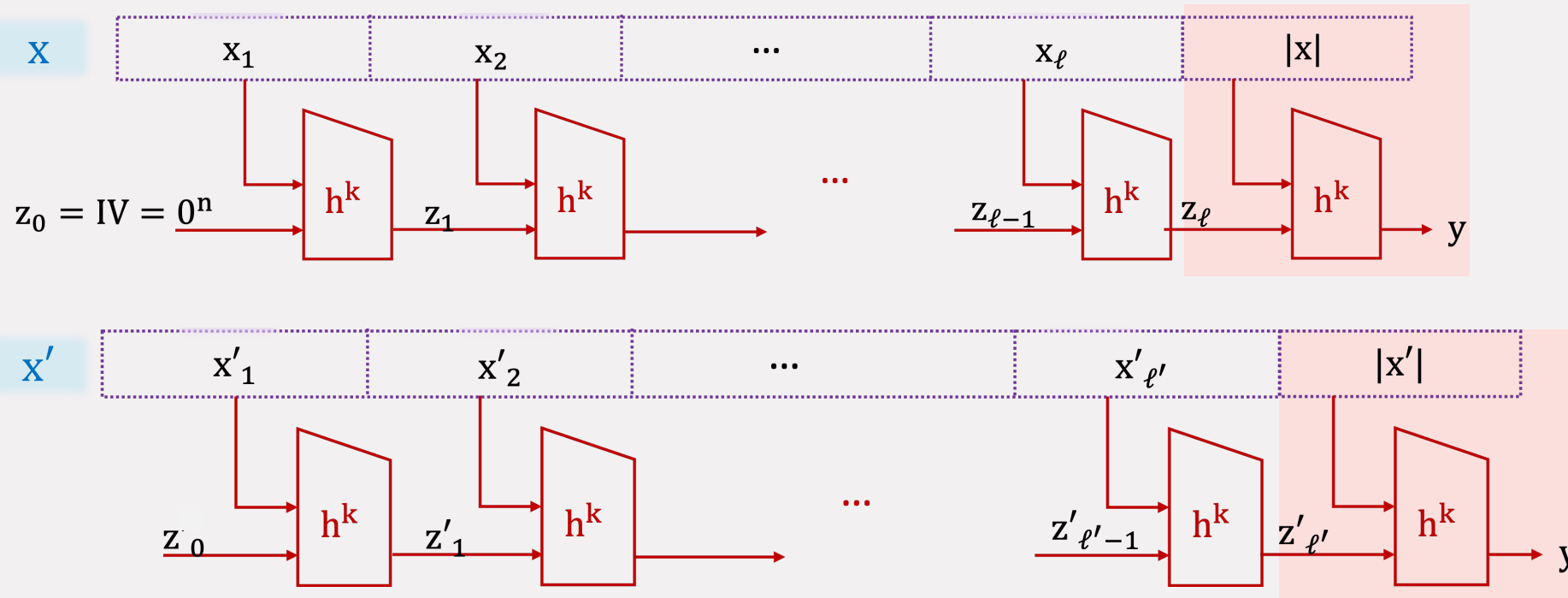
Merkle Damgård Transform : Proof Idea

If you have a collision x, x' for H^k

Case 1: x, x' are of different lengths ($|x| \neq |x'|$)

$$h^k(z_\ell || |x|) = h^k(z'_{\ell'} || |x'|) = y$$

the collision for h^k



Merkle Damgård Transform : Proof Idea

If you have a collision x, x' for H^k

Case 1: x, x' are of different lengths ($|x| \neq |x'|$)

$$h^k(z_\ell || |x|) = h^k(z'_{\ell'} || |x'|) = y$$

the collision for h^k

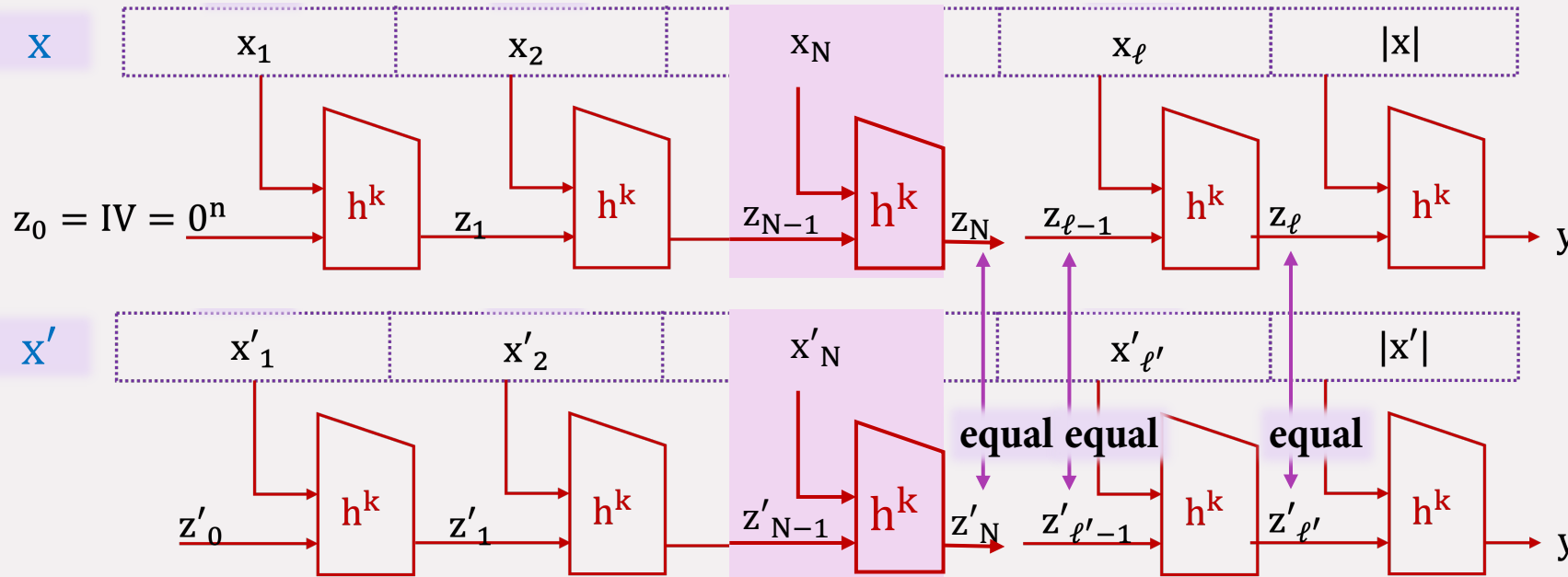
Case 2: x, x' are of same length ($|x| = |x'|$)

$i = N$ be the largest index for which $z_{i-1} || x_i \neq z'_{i-1} || x'_i$

the collision for h^k

(Since $x \neq x'$ and $|x| = |x'|$
such an N exists)

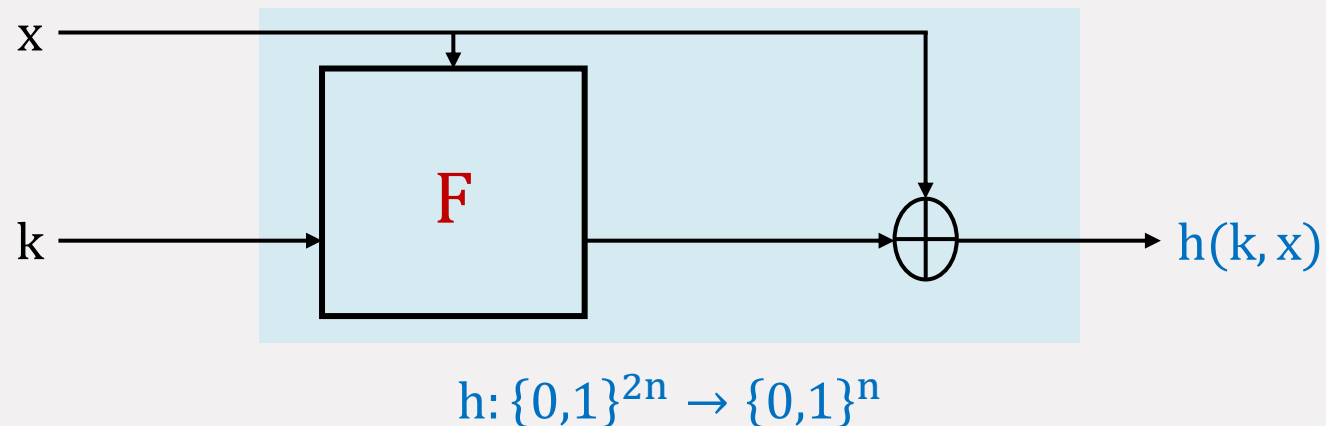
$$\Rightarrow z_N || x_{N+1} = z'_N || x'_{N+1}$$



Example of a Compression Function

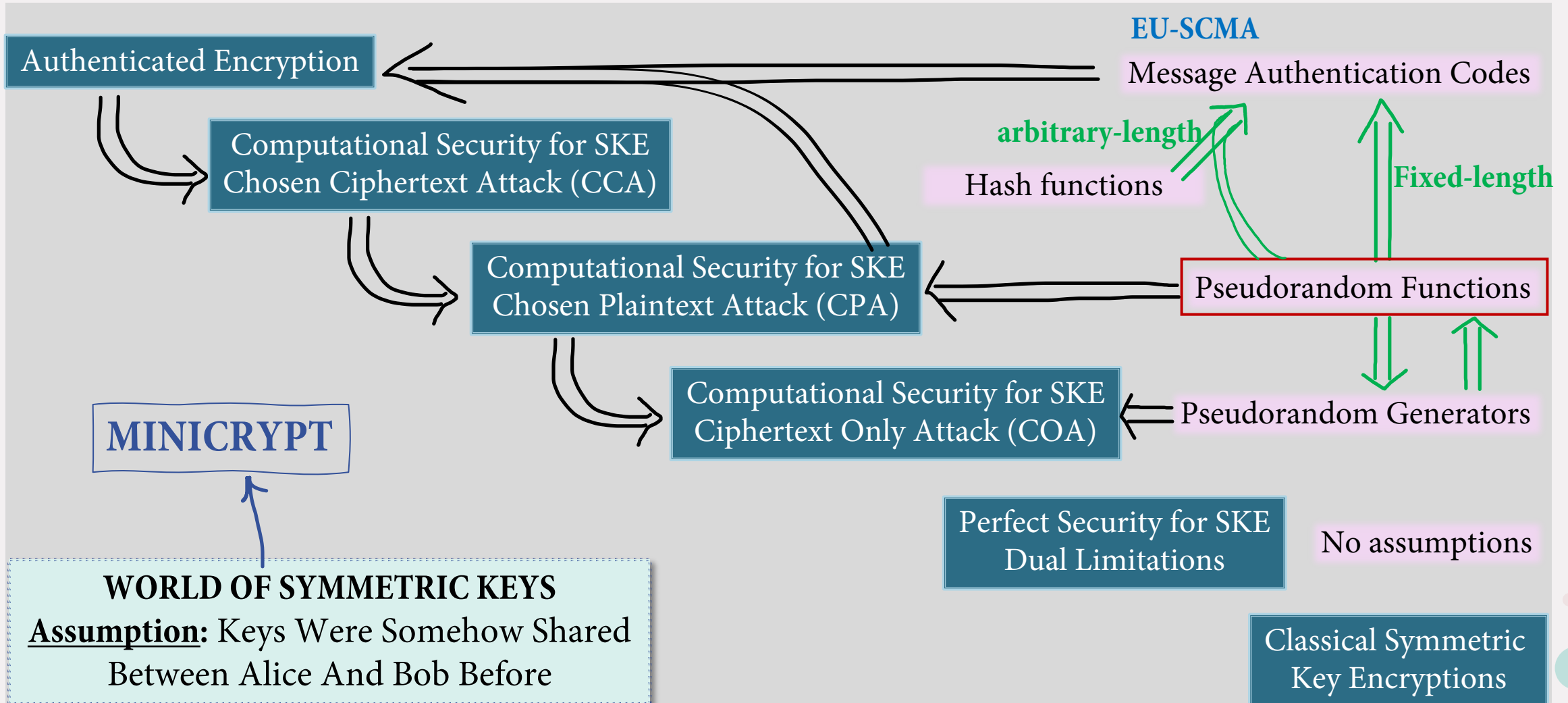
Davies-Meyer Construction

F be a block cipher with n -bit keys and n -bit input length

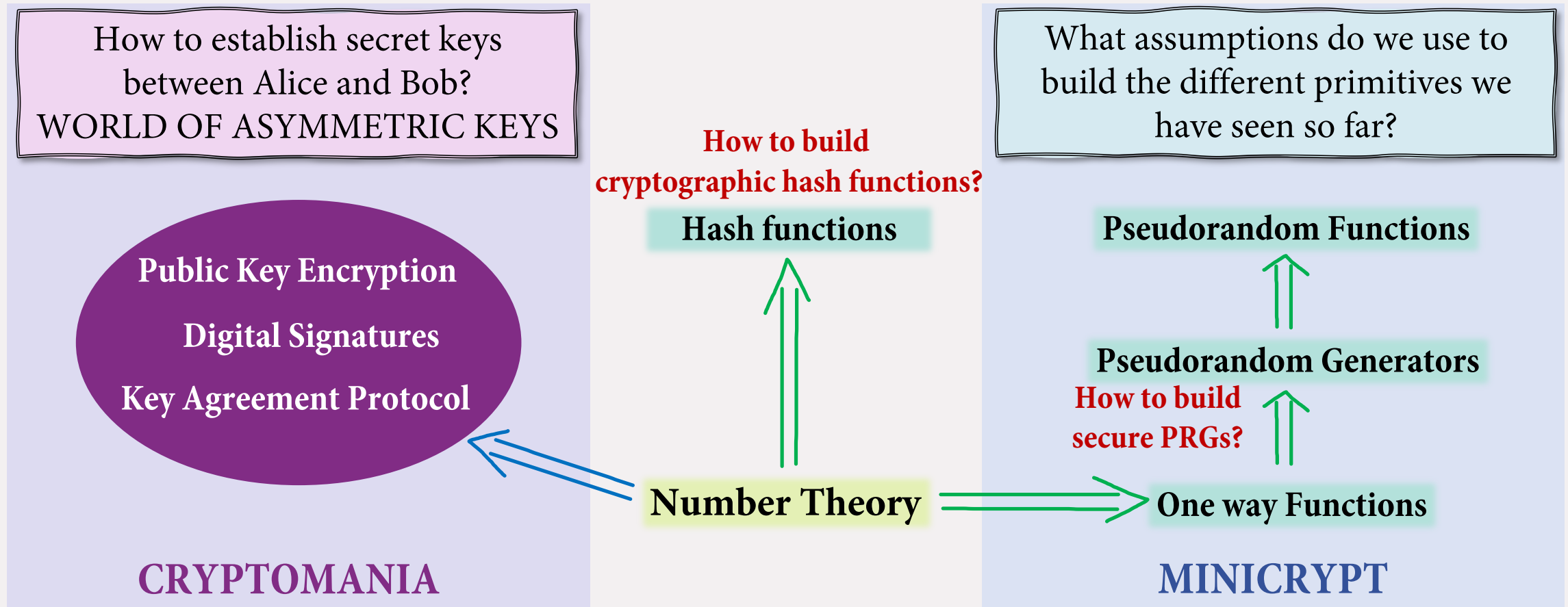


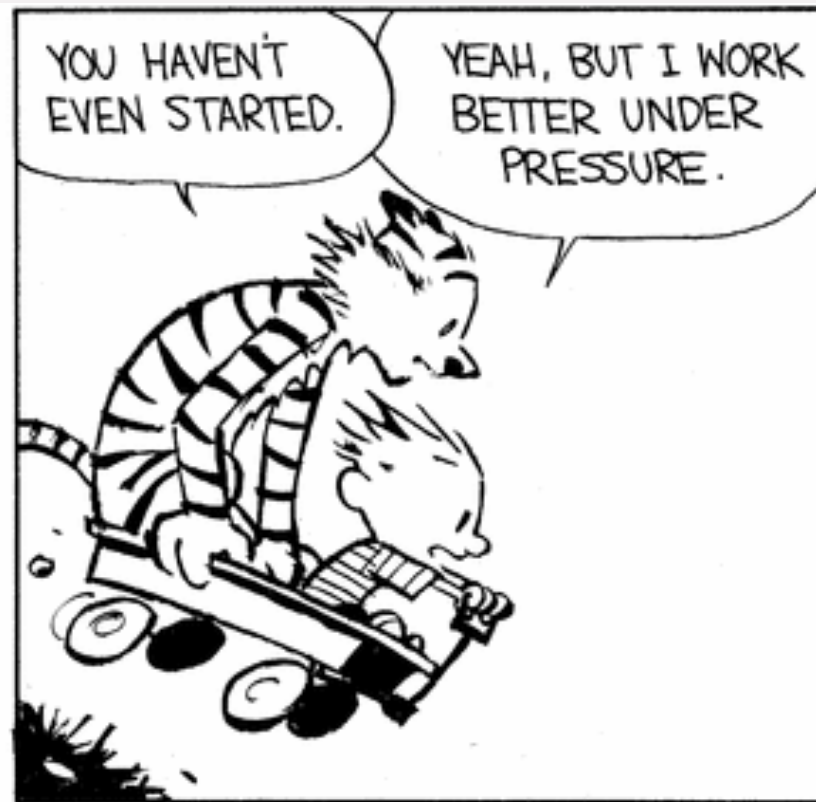
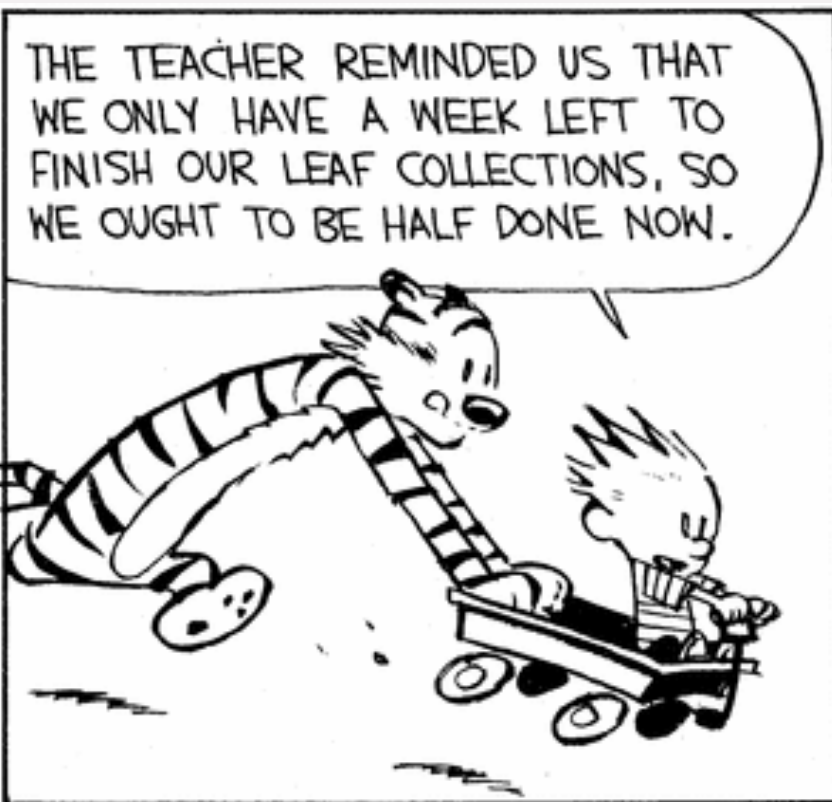
- There is no known proof that Davies-Meyer is collision resistant, just assuming an SPRP.
- To prove its security, need to assume a strong assumption about F (*“ideal cipher”*).
- Hard to instantiate F (any block cipher cannot be plugged in naively!)
(E.g. All SHA-family is constructed using Davies Meyer + Merkle Damgård)

Journey so far



Journey Ahead: After MidSem





THANK YOU